Final Project
Eigen-value Decomposition by Iterative QR Operations

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1 Hardware Parameters

To determine the optimal word-length that satisfies the Root Mean Square Error (RMSE) constraint (RMSE $< 10^{-2}$) for both eigenvalues and eigenvectors while minimizing hardware overhead, a systematic hardware-algorithm co-design approach was adopted. The optimization focuses on balancing the fractional word-length (W_f) and the number of CORDIC stages (S), as they govern the quantization error and the angular resolution error, respectively.

1.1 Theoretical Analysis of the Area-Time (AT) Product

- **Fractional Word-length (W_f):** Reducing W_f linearly decreases the hardware area by narrowing the datapath (i.e., adders, registers, and multiplexers). It also slightly improves the critical path delay.
- **CORDIC Stages (S):** Reducing S provides a dual benefit to the Area-Time (AT) product. It eliminates entire unrolled pipeline stages (drastically reducing area) and decreases the overall pipeline latency (reducing total execution cycles).
- **The Golden Rule ($S \approx W_f$):** It is mathematically redundant to set $S \gg W_f$. In fixed-point arithmetic, an arithmetic right shift by $i \geq W_f$ during the i-th CORDIC stage results in zero. Thus, any stages beyond W_f simply waste hardware without improving accuracy.

1.2 Two-Step Parameter Search Methodology

To discover the optimal specifications, a Python-based bit-true model was developed to sweep the parameters systematically:

1. **Isolating Quantization Error:** The stage count S was initially set to a sufficiently large value ($S = 16$) to practically eliminate angular error. By sweeping W_f, the minimum fractional word-length required to keep the RMSE strictly below the 10^{-2} threshold was identified.
2. **Minimizing Algorithmic Error:** Fixing W_f to the discovered minimum, S was subsequently swept downwards. The optimal S was chosen as the minimum value right before the RMSE diverges and violates the 10^{-2} constraint.

1.3 Fine-tuning and Hardware Optimizations

After obtaining the baseline (W_f, S), further fine-tuning was performed jointly on S, W_f, and the fractional bits for the CSD (Canonical Signed Digit) scaling factor to finalize the most resource-efficient combination.

A crucial architectural optimization was substituting standard truncation with **rounding to the nearest integer (Round Half Up)**. Truncation introduces a constant negative DC bias that accumulates significantly over the dense CORDIC iterations required for the QR algorithm. By implementing rounding—which incurs near-zero area overhead by simply adding the discarded MSB to the carry-in of the respective adders—the accumulated bias was virtually eliminated. This optimization enabled the design to strictly meet the 10^{-2} accuracy target using an even smaller W_f, yielding significant area savings while ensuring numerical stability.

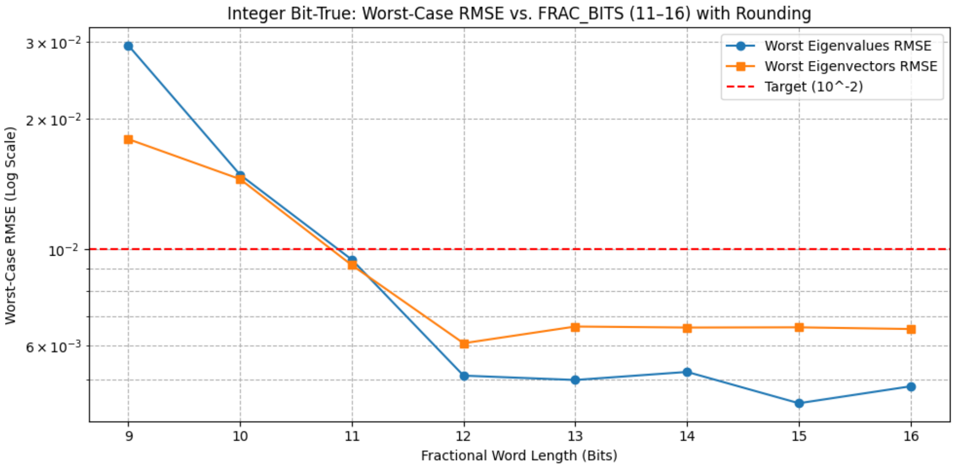

Figure 1: RMSE vs. Fractional Word-length

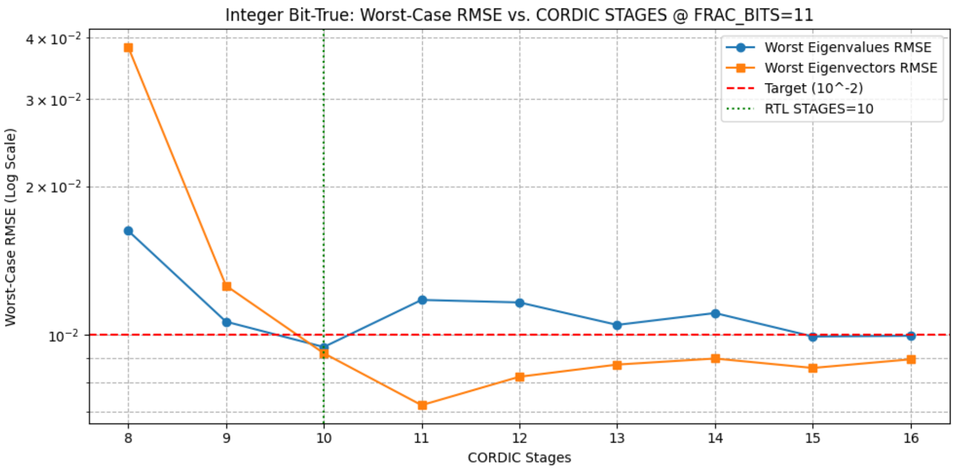

Figure 2: RMSE vs. CORDIC Stages

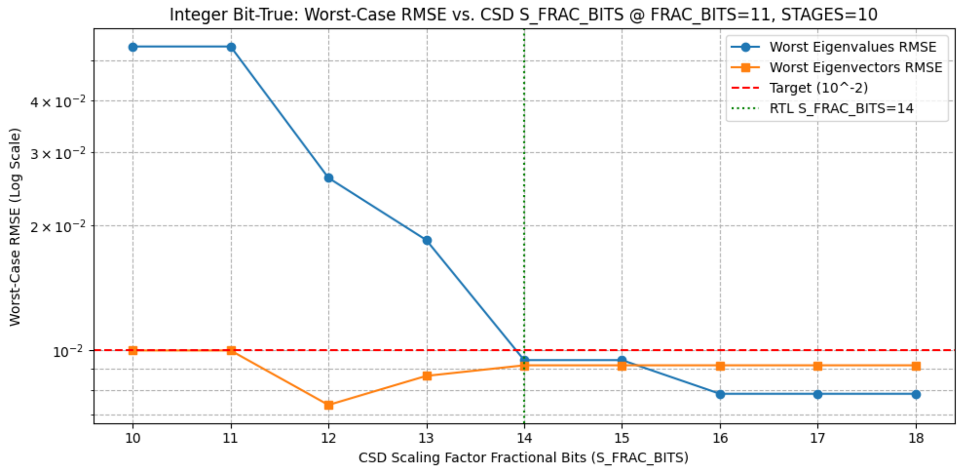

Figure 3: RMSE vs. CSD Word-length

Base on Figure 1, 2, and 3, the optimal parameters are:

- Fractional Word-length (W_f): 11
- CORDIC Stages (S): 10
- CSD Word-length (W_{CSD}): 14

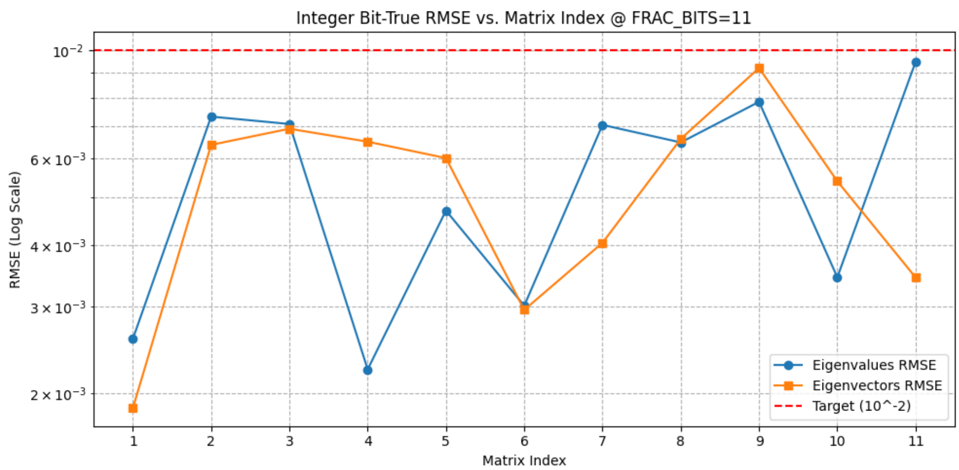

Figure 4: RMSE vs. Matrix Indices

2 Hardware Design Concept

2.1 Overall Architecture Overview

The proposed hardware architecture is a highly optimized, fully pipelined 3-PE (Processing Element) CORDIC-based systolic array designed to compute the Eigenvalue Decomposition (EVD) of a 3×3 symmetric matrix using the iterative QR algorithm. To minimize the Area-Time (AT) product while satisfying the stringent 10^{-2} Root Mean Square Error (RMSE) constraint, the datapath operates on a tailored fixed-point Q6.11 format (17-bit total word-length) with 10 CORDIC micro-rotation stages.

A critical architectural novelty in this design is the integration of an **Optimized Zero-Adder Rounding** logic paired with **Adder Tree Separation** in the final CSD (Canonical

Signed Digit) scaling stage. This removes the massive 12-operand addition bottleneck by isolating the 1-bit rounding constants into a miniature 5-bit prefix adder, successfully reducing the critical path logic depth and ensuring timing closure under high-frequency synthesis conditions.

2.2 Transpose Buffer Design and Forced Symmetry

The iterative QR algorithm necessitates updating the matrix via $T_{next} = R \cdot Q$ and $U_{next} = U \cdot Q$. In a hardware systolic array, this matrix multiplication is achieved by feeding the transposed matrices, R^T and U^T, back into the CORDIC array operating in rotation mode.

Instead of deploying a full 3×3 (9-register) buffer for the T matrix, the design implements an **Optimized 6-Register Transpose Buffer**. Due to finite word-length effects (truncation and rounding errors) over deep CORDIC pipelines, the theoretical Hermitian symmetry of the T matrix is inherently degraded over multiple iterations. By exclusively capturing the lower-triangular elements ($M_{11}, M_{12}, M_{13}, M_{22}, M_{23}, M_{33}$) and structurally mirroring them during the feedback routing (e.g., routing M_{12} as both T_{12} and T_{21}), the hardware explicitly forces perfect Hermitian symmetry. This area-saving technique intrinsically acts as a numerical stabilizer, preventing the algorithm from diverging due to accumulated asymmetric quantization errors.

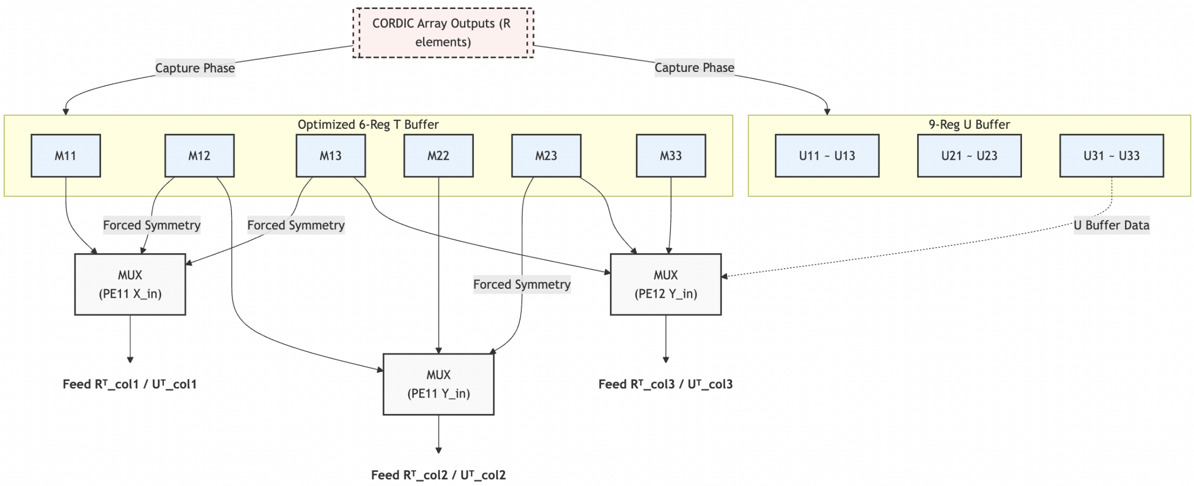

Figure 5: Block Diagram of the Transpose Buffer illustrating Forced Symmetry Multiplexing.

Note on U Buffer Routing Representation: In the actual hardware implementation, the entire 9-register U buffer is fully connected to all three multiplexers (PE11 X_{in}, PE11 Y_{in}, and PE12 Y_{in}) to stream the transposed matrix U^T column-by-column during the rotation phase. However, for visual clarity, this comprehensive data bus is symbolically abstracted as a single dashed arrow. This simplification prevents severe wire clutter and ensures that the reader's attention remains focused on the primary architectural novelty of this design: the intricate cross-routing logic of the forced Hermitian symmetry within the optimized 6-register T buffer.

2.3 Iterative Operation Control Logic

The sequential execution of the iterative EVD algorithm is governed by a global Finite State Machine (FSM) utilizing a multi-level counter hierarchy: a macro-level `iter_cnt` for algorithm iterations and a micro-level `phase_timer` for pipeline cycle control. The operational flow is explicitly defined by four operational states:

1. **ST_IDLE:** The system initializes the transpose buffer, loading the 3×3 identity matrix

into the U_{buf}. It remains in this state until the `InValid` signal is asserted, triggering the streaming of the initial symmetric matrix into the input column registers.

2. **ST_QR (Vectoring Phase):** The CORDIC array is configured to vectoring mode (`mode = 1`). The feed controller streams the columns of the input matrix (or T_{next} from the previous iteration) into the systolic PEs. During this phase, each PE computes the required rotation angles to eliminate sub-diagonal elements and securely latches the CCW/CW decisions into internal `d_latch` shift registers. Upon reaching the predefined pipeline latency cycles, the resultant upper-triangular matrix R is captured into the 6-register M_{buf}.

3. **ST_ROT_PIPE (Rotation Phase):** The CORDIC PEs are switched to rotation mode (`mode = 0`). To maximize hardware utilization, this phase is highly pipelined. The feed controller acts as a dynamic transpose network, reading data from M_{buf} and U_{buf} in a transposed sequence. It first feeds R^T into the array to compute $T_{next} = R{\cdot}Q$, immediately followed by U^T (with a 3-cycle offset) to compute $U_{next} = U \cdot Q$. The PEs reuse the exact micro-rotation sequences stored in `d_latch` from the preceding `ST_QR` phase. The pipelined results are seamlessly overwritten into M_{buf} and U_{buf} as they emerge from the array.

4. **ST_DONE:** Once the macro counter `iter_cnt` reaches the target limit ($ITER\_MAX = 7$), the FSM transitions to the completion state. It sequentially streams out the stable Eigenvalues (extracted directly from the diagonal registers M_{11}, M_{22}, M_{33}) followed by the structured Eigenvectors from the U_{buf} across four consecutive valid clock cycles.

3 Hardware Simulation Results

Figure 6 and 7 show the behavior simulation and post-synthesis simulation results with the input patterns quantized from Matrix(:,:,8).

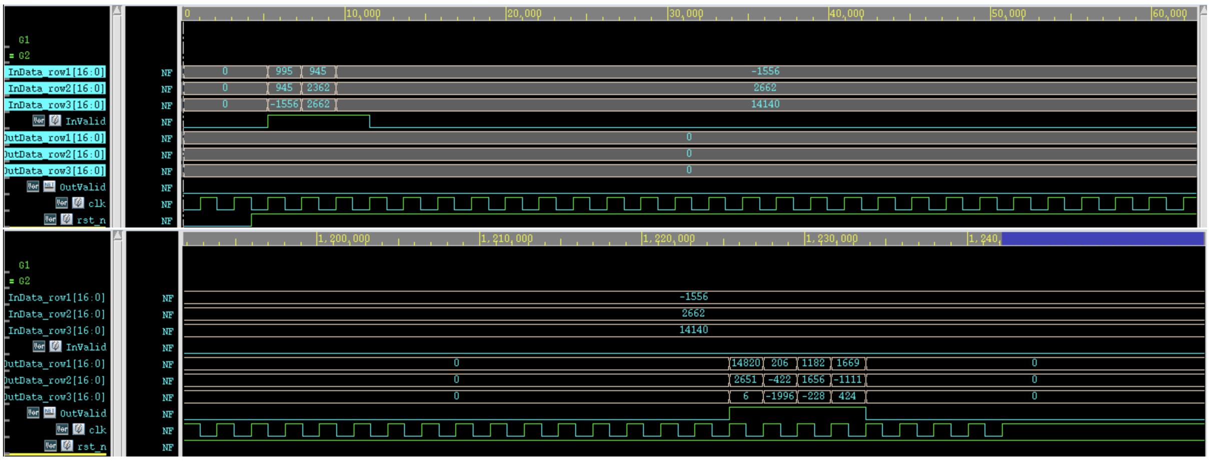

Figure 6: Behavior simulation timing waveform. The waveform between InValid and OutValid is skipped since no valid output data is produced during this period.

The integer eigenvalues are (14820, 2651, 6), converted to fixed-point Q6.11 format are (14820/2048, 2651/2048, 6/2048) = (7.236328125, 1.2939453125, 0.0029306640625). Compared to the eigenvalues calculated by `np.linalg.eigh` from NumPy (7.24685273, 1.29717242, 1.52791938e-04), the RMSE is 6.4802e-03, which is within the tolerance of 10^{-2}.

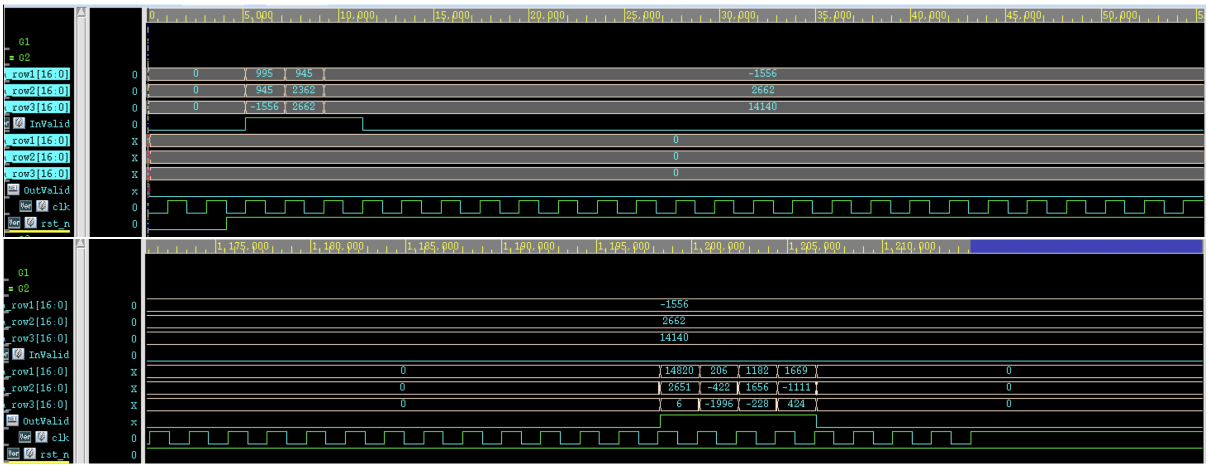

Figure 7: Post-synthesis simulation timing waveform. The waveform between InValid and OutValid is skipped since no valid output data is produced during this period. Clock period $T = 0.205$ ns.